HistoAtlas: an open, interpretable pan-cancer histomics resource linking tissue morphology to molecular programs and clinical outcomes

Pierre-Antoine Bannier[a,*]

[a]*Independent researcher*

Summary

Routine hematoxylin-and-eosin pathology is the most widely collected tissue assay in oncology, yet quantitative morphology is rarely released as reusable data. HistoAtlas converts 6,745 diagnostic whole-slide images from 6,088 patients across 21 cancer types into 38 named, compartment-resolved morphological measurements and links each to survival, gene expression, copy number, pathway activity and immune state, yielding 487,638 precomputed correlations and 5,623 survival models reported with effect sizes, confidence intervals and declared multiple-testing families. Six sparse programs summarize the measurements. In 1,095 slides from 817 patients profiled by an independent consortium, the measurements and their transcriptomic links transferred and extended to protein, with a narrow residual that survives adjustment for matched transcripts. HistoAtlas is searchable through a web atlas and a programmatic interface, and every statistic traces back to tissue compartments and individual cells.

Keywords: computational pathology, histomics, pan-cancer atlas, interpretable representations, multimodal data integration, tumor microenvironment, external validation, FAIR data

*Lead contact and corresponding author.1

Email address: pierreantoine.bannier@gmail.com (Pierre-Antoine Bannier)

Highlights

- 6,745 pathology slides become 38 named, compartment-resolved measurements
- Morphology-molecule links replicate in 1,095 slides from a second consortium
- Six sparse morphology programs map to pathways and to protein abundance
- An open atlas exposes 487,638 precomputed correlations to people and agents

The bigger picture

Hematoxylin-and-eosin (H&E) stained tissue sections are prepared for nearly every solid tumor diagnosed, and public cohorts such as The Cancer Genome Atlas pair these slides with genomic, transcriptomic and clinical data. The tissue features a pathologist assesses, such as tumor content, the position of immune cells relative to tumor epithelium, nuclear size and shape, or mitotic activity, are nevertheless rarely recorded quantitatively. Without such measurements, slide images cannot be analyzed jointly with molecular and clinical data using standard statistical methods.

HistoAtlas defines 38 quantitative tissue measurements, each computed from segmented tissue compartments and classified cells by a fixed formula with a physical unit, and applies them to 6,745 slides from 21 cancer types. Every measurement is tested against survival, gene expression, copy number, pathway activity and immune state within a single statistical framework: effect sizes with bootstrap confidence intervals, false discovery rate control within declared families of tests, minimum detectable effects at fixed power, and proportional-hazards checks for survival models. Because each value is computed from identified cells in identified tissue regions, any association can be inspected at the level of the image tiles that produced it. The measurements and their molecular associations were re-evaluated, without refitting, in an independent cohort of 1,095 slides, where most prespecified associations replicated. The complete tables are released through a web atlas and a programmatic interface, so that new hypotheses can be checked against existing results without reprocessing the images. The same approach, in which expert visual assessments are defined as explicit measurements, their uncertainty is quantified and their transfer across cohorts is tested, could be applied to other imaging data that are currently interpreted qualitatively by specialists.

1. Introduction

Hematoxylin-and-eosin (H&E) staining of tissue sections is the most widely used assay in cancer medicine [1]. Every diagnostic slide records tissue composition, cell densities, nuclear morphology and the spatial arrangement of tumor and microenvironmental cells [2, 3]. For genomics, transcriptomics and proteomics, pan-cancer resources release standardized, queryable tables [4, 5, 6, 7]; no equivalent resource exists for histopathology. Diagnostic slides are archived at scale, but the quantities a pathologist assesses from them are rarely recorded as numerical data.

Bulk molecular assays average over the whole sample and do not record the spatial organization of the tissue. Yet the density of immune cells in the tumor core and at the invasive margin is prognostic in several cancer types [8, 9, 10]. Saltz et al. showed that deep learning applied to public pathology slides recovers tumor-infiltrating lymphocyte density at pan-cancer scale [11], but a density computed over the whole slide does not distinguish lymphocytes inside tumor epithelium from lymphocytes confined to the surrounding stroma, the pattern that defines immune exclusion. Existing public resources each cover part of the problem: cBioPortal provides molecular and clinical tables without quantitative morphology [12], the Cancer Imaging Archive provides images without associated statistical analyses [13], and the Human Protein Atlas provides immunohistochemistry-based maps of individual proteins [14, 15], which are not available for routine H&E slides.

Computational pathology has made the extraction of tissue features tractable. Early morphometric studies linked handcrafted image features to outcome and molecular alterations in single cancer types [16, 17, 18]; more

recent systems predict molecular alterations, expression programs and outcomes directly from H&E [19, 20, 21], as reviewed in [22]; and foundation models now produce tile-level embeddings that can be aggregated to the slide level [23, 24, 25]. These results show that H&E images contain substantial biological information. Foundation-model embeddings are information-dense representations that support many downstream tasks, including mutation, biomarker and outcome prediction [26, 27, 21, 28], but they are not interpretable: an individual coordinate does not correspond to a defined tissue property, so an association found with an embedding cannot be attributed to specific cells or tissue regions. Interpretable feature sets have been proposed [29, 30], but no pan-cancer set of interpretable measurements has been released together with its molecular and clinical associations and evaluated in an independent cohort.

Image statistics depend on staining protocol, scanner, fixation and case selection, so a measurement defined in one cohort can have a different distribution, and a different relationship to molecular variables, in another. Measurements and associations derived from The Cancer Genome Atlas (TCGA) therefore need to be re-estimated on slides from other institutions. The Clinical Proteomic Tumor Analysis Consortium (CPTAC) provides independent diagnostic slides with matched transcriptomic and proteomic profiles across several cancer types [6]. This allows two tests: whether the measurement distributions and their correlations with gene expression are reproduced in an independent cohort, and whether the same correlations are observed at the protein level.

Here we present HistoAtlas. From 6,745 TCGA diagnostic slides from

6,088 patients spanning 21 solid-tumor types, we derive 38 interpretable, compartment-resolved histomic measurements and compute their associations with survival, gene expression, copy number, pathway activity and immune state, reporting effect sizes, confidence intervals, minimum detectable effects and explicitly declared multiple-testing families throughout. We show that the 38 measurements can be summarized by six continuous programs, each correlated with specific MSigDB Hallmark pathways [31]; that the measurement distributions, their correlations with gene expression and two of the six programs are reproduced in 1,095 independent CPTAC slides; that the correlations with gene expression are also observed at the protein level, with 132 measurement–protein correlations remaining significant after adjustment for the matched transcript; and that adding the measurements to survival models that already include clinical covariates and matched transcripts slightly improves discrimination. The resource is released as an interactive atlas and a programmatic interface, in which every reported statistic is linked to the tissue compartments and individual cells from which it was computed.

2. Results

2.1. HistoAtlas turns routine pathology into a named, compartment-resolved measurement layer

We built HistoAtlas from 6,745 H&E-stained diagnostic whole-slide images from 6,088 patients spanning 21 TCGA solid-tumor types (Table S1). Twelve further TCGA types were excluded because their dominant cell morphologies fall outside the training domain of the cell detection model, including lymphoid, glial, melanocytic, mesenchymal, neuroendocrine, renal tubular and germ-cell

tumors (Table S2). Descriptive analyses of the measurement space use every slide; all statistical associations use one representative slide per patient, so that the 312 patients with several slides are not counted more than once.

The pipeline converts each slide into quantitative morphology in two stages (Figure 1A). A tissue model assigns every region to one of five compartments (cancerous epithelium, stroma, necrosis, normal epithelium, blood); tumor and stroma together account for more than 90% of analyzed tissue area, so downstream measurements can be expressed relative to biologically meaningful territory rather than to whole-slide area. The HistoPLUS cell model [32], which classifies cells into 13 types, then detects individual cells, of which nine types are used here (tumor cell, lymphocyte, fibroblast, plasmocyte, neutrophil, eosinophil, red blood cell, apoptotic body, mitotic figure), more than four billion cells across the cohort. Signed distance transforms from the tumor boundary define spatial bands, so that a lymphocyte counted inside tumor epithelium is not confused with one counted 150 µm away in stroma. From these maps we derived 38 histomic measurements in six groups: tissue composition (3), cell densities (8), nuclear morphology and kinetics (7), spatial organization (11), spatial heterogeneity (5) and cell-type ratios (4) (Table S3).

Across 22 analysis cohorts (21 cancer types plus a pan-cancer set) we fitted 5,623 Cox proportional hazards models (four endpoints, each under an unadjusted and a covariate-adjusted specification), of which 5,283 yielded valid estimates after proportional-hazards screening, and computed 487,638 Spearman correlations against messenger RNA expression, pathway activity, copy number and immune deconvolution scores. Benjamini–Hochberg

correction was applied within declared, non-overlapping families (150 for correlations, defined by cancer type, molecular data type and model). At a false discovery rate of 0.05, 260 survival associations and 88,920 molecular correlations were significant (87,217 of them involving the 38 measurements; the others involve three quality-control variables released in the same table; Table S8). Every association carries an effect size, a bootstrap 95% confidence interval, raw and adjusted P values, a minimum detectable effect at 80% power, a proportional-hazards flag where applicable, and the identifier and size of its correction family.

Pairwise Spearman correlation across the 38 measurements resolves coherent modules: immune-density measurements form a tight positive block, nuclear morphometry clusters separately, and cross-module anticorrelations, such as between tumor area fraction and the tumor–stroma interface measurements, delineate distinct axes (Figure 1B). Principal component analysis gives an effective dimensionality of 18 components at 90% explained variance, so the layer is compressible but not collinear. A UMAP embedding of all 6,745 slides recovers expected morphological neighborhoods, with breast, liver, thyroid and endometrial carcinoma occupying distinct regions and squamous and adenocarcinoma types interleaved in the center (Figure 1C). Unsupervised K-means partitioning into ten pan-cancer clusters recovers groups whose cancer-type composition and measurement profiles are interpretable without any molecular input (Figure 1D,E), and which are enriched for distinct immune subtypes and Hallmark pathway states when molecular annotation is subsequently overlaid (Figure S1). Two of these purely morphological groups separate prognostically in Cox models stratified by cancer type, so the effects

are not restatements of tissue of origin: a low-proliferation cluster dominated by liver, thyroid and adrenocortical carcinoma (C2) was associated with better overall survival (HR $= 0.54$, 95% CI $[0.40, 0.73]$, $P_{\text{adj}} = 6.3 \times 10^{-4}$; 516 patients in the cluster, 108 events), and a high-interface cluster dominated by breast and prostate carcinoma (C8) with worse survival (HR $= 1.36$, 95% CI $[1.15, 1.62]$, $P_{\text{adj}} = 1.8 \times 10^{-3}$; 1,112 patients, 214 events).

2.2. Six continuous morphological programs summarize the measurement space

We applied sparse principal component analysis ($\alpha = 1.0$) to the 38 measurements across 5,371 TCGA patients with complete measurement vectors, yielding six continuous programs, each loading on 29 to 33 measurements with a few dominant ones (Figure 2A). Their biological identities were not assigned in advance but read off from correlations with 50 MSigDB Hallmark pathway scores [31] (Figure 2C): program 1 correlates most strongly with ALLOGRAFT_REJECTION ($\rho = 0.59$, $n = 5{,}371$) and the interferon-γ response ($\rho = 0.51$), program 2 with ESTROGEN_RESPONSE_EARLY ($\rho = 0.44$) and EPITHELIAL_MESENCHYMAL_TRANSITION ($\rho = 0.41$), and program 3 with MYC_TARGETS_V1 and G2M_CHECKPOINT (both $\rho = 0.53$). We therefore label them immune infiltration, stromal architecture, nuclear atypia and proliferation, tumor density, myeloid inflammation and fibroblast biology. The same correspondence is visible at the level of individual measurements, where immune densities align with immune pathway signatures, nuclear morphometry and mitotic index with cell cycle and proliferation signatures, and stromal and interface measurements with the epithelial–mesenchymal transition, angiogenesis and coagulation signatures (Figure S2A).

2.3. Compartment-resolved immune topology is prognostic and molecularly anchored

Tumor-infiltrating lymphocytes are conventionally scored in the stromal compartment [33]; HistoAtlas measures lymphocyte density separately in tumor epithelium and in stroma. Intratumoral lymphocyte density was associated with improved overall survival in the pan-cancer adjusted model (hazard ratio [HR] = 0.87, 95% CI [0.81, 0.93], $P_{\text{adj}} = 9.8 \times 10^{-4}$, n = 4,560, 1,356 events), with the strongest cancer-specific effects in breast carcinoma (HR = 0.72, 95% CI [0.60, 0.88], $P_{\text{adj}} = 0.018$, $n = 960$) and head-and-neck squamous carcinoma (HR = 0.74, 95% CI [0.63, 0.87], $P_{\text{adj}} = 3.9 \times 10^{-3}$, $n = 444$; Figure 3A,B). Stromal lymphocyte density, measured on the same slides by the same model, showed a weaker pan-cancer effect (HR = 0.89, 95% CI [0.83, 0.97], $P_{\text{adj}} = 0.031$, n = 4,561) and no significant effect in breast carcinoma (HR = 0.93, 95% CI [0.77, 1.12], $P_{\text{adj}} = 0.67$). The two compartments therefore separate: localization within tumor epithelium, not lymphocyte abundance alone, carries the stronger prognostic signal, consistent with the prognostic value of immune-cell location reported in colorectal cancer [8, 10]. In most of the 18 cancer types with an evaluable adjusted model the intervals are wide and include 1, which is the expected behavior of small cohorts (Figure S3B).

The same distinction appears in the molecular data. Tumor–lymphocyte nearest-neighbor distance at the invasive front, a direct spatial measure of immune exclusion [34], was inversely associated with *CD8A* expression in breast carcinoma ($\rho = -0.53$, 95% CI [−0.58, −0.48], $P_{\text{adj}} = 1.8 \times 10^{-68}$, $n = 958$; Figure 3C), and the top gene correlates of intratumoral lymphocyte

density are dominated by T-cell, B-cell and checkpoint transcripts (*TIGIT*, *CD3E*, *CD3D*, *CD79A*, *PDCD1*, *CD8A*, *CTLA4*), with luminal markers (*ESR1*, *PGR*, *CCND1*) among the strongest negative correlates (Figure 3D).

2.4. Named measurements recover proliferation, invasion and differentiation programs

In TCGA breast carcinoma, mitotic index correlated with canonical cell-cycle regulators, including *CCNE1* ($\rho = 0.568$, 95% CI $[0.519, 0.617]$, $P_{\text{adj}} = 2.7 \times 10^{-80}$, $n = 958$), *PLK1* ($\rho = 0.558$ $[0.510, 0.607]$, $P_{\text{adj}} = 5.1 \times 10^{-77}$), *AURKA* ($\rho = 0.539$, $P_{\text{adj}} = 4.7 \times 10^{-71}$), *CHEK1* ($\rho = 0.526$) and *BIRC5* ($\rho = 0.520$; all $P_{\text{adj}} < 10^{-64}$). Invasion depth, defined as the 75th percentile of tumor distance from the tissue boundary, aligned with an epithelial–mesenchymal axis (*ZEB1*: $\rho = 0.318$ $[0.262, 0.375]$, $P_{\text{adj}} = 8.4 \times 10^{-23}$, $n = 957$; *ALDH1A1*: $\rho = 0.327$, $P_{\text{adj}} = 3.8 \times 10^{-24}$). Nuclear pleomorphism, the coefficient of variation of tumor nuclear area, correlated positively with proliferation genes (*PLK1*: $\rho = 0.484$, $P_{\text{adj}} = 1.4 \times 10^{-55}$) and negatively with luminal differentiation markers (*BCL2*: $\rho = -0.374$, $P_{\text{adj}} = 8.1 \times 10^{-32}$; *ESR1*: $\rho = -0.358$, $P_{\text{adj}} = 4.4 \times 10^{-29}$). These correspondences restate, in continuous and automatically measured form, relationships that underpin histopathological grading [35]. Their value here is that they are computed identically on every slide in the atlas, so they can be compared across 21 cancer types rather than asserted one tumor type at a time. Across all molecular data types the coupling is strongest for expression and pathway activity and weakest for copy number: 80% of pan-cancer adjusted expression correlations and 83% of pathway correlations were significant, against 52% for copy number (Figure S2B).

2.5. The measurement layer and its molecular links transfer to an independent consortium

Portability was assessed on 1,095 CPTAC slides from 817 patients across five cancer types with TCGA counterparts: breast ($n = 198$ slides), lung adenocarcinoma ($n = 240$), head-and-neck squamous carcinoma (390 slides from 112 patients), pancreatic adenocarcinoma ($n = 175$) and endometrial carcinoma ($n = 92$).

For each cancer type we computed the standardized difference between cohort medians, in units of the TCGA standard deviation, for each measurement, with a prespecified criterion of at least 31 of 38 measurements within one standard deviation. Four of five types passed: lung adenocarcinoma and pancreatic adenocarcinoma at 34/38 (89.5%), head-and-neck at 32/38 (84.2%) and breast at 31/38 (81.6%); endometrial carcinoma narrowly missed at 30/38 (78.9%), driven by tissue-composition shifts in the smallest cohort ($n = 92$ versus $n = 459$ in TCGA; Figure 4A; Table S9). The inter-measurement correlation geometry was preserved across cohorts (Pearson $r = 0.84$ over the 703 upper-triangle pairs, $P < 10^{-180}$; Table S9), indicating that the 38 measurements span the same latent morphological space in independent tissue.

The morphology-to-transcriptome bridge replicated. All ten prespecified histomic–molecular associations matched in direction and nine remained significant after correction within the full CPTAC family of approximately 24,700 tests per model per cancer type, including mitotic index with *PLK1* in lung adenocarcinoma ($\rho = 0.505$, $P_{\text{adj}} = 5.0 \times 10^{-5}$), invasion depth with *ZEB1* in breast carcinoma ($\rho = 0.319$, $P_{\text{adj}} = 4.6 \times 10^{-3}$) and nuclear

pleomorphism with *ESR1* ($\rho = -0.389$, $P_{\text{adj}} = 2.7 \times 10^{-4}$) and *BCL2* ($\rho = -0.416$, $P_{\text{adj}} = 7.7 \times 10^{-5}$). Beyond the prespecified set, effect sizes were concordant at scale. Across 750 matched feature–gene–cancer triples drawn from each cohort's strongest correlates, the two cohorts' Spearman coefficients correlated at Pearson $r = 0.787$ (95% CI $[0.758, 0.813]$, $P = 4.0 \times 10^{-159}$), with mean $|\rho|$ modestly attenuated in CPTAC (0.251 versus 0.301). Across all 25,155 evaluable measurement–gene pairs, most of which are near zero in both cohorts, the correlation was $r = 0.56$ (Figure 4B), as expected when weak associations are estimated in small cohorts.

Program transfer was mixed in an informative way. Projecting the six TCGA-fitted programs into 297 CPTAC patients with complete measurement vectors, two-sample Kolmogorov–Smirnov tests showed no significant distributional shift for the immune program ($D = 0.072$, $P = 0.10$) or the fibroblast program ($D = 0.064$, $P = 0.19$), a borderline shift for the stromal program ($D = 0.096$, $P = 0.010$), and larger shifts for the remaining three ($D = 0.15$ to 0.34, $P < 10^{-4}$; Figure 2B). The programs also validated at the protein level: the immune program tracked CXCL9 ($\rho = 0.85$), CD79A ($\rho = 0.82$) and CD3E ($\rho = 0.78$) protein abundance, and the stromal program tracked ZEB1 ($\rho = 0.72$) and FN1 ($\rho = 0.71$), across 2,220 program–protein pairs in four cancer types (Table S10). By contrast, transferring the ten hard cluster labels by nearest-centroid assignment reproduced pathway enrichment only weakly (Spearman $\rho = 0.15$, $P = 1.4 \times 10^{-3}$ across 450 cluster–pathway pairs), with three of nine evaluable clusters concordant ($\rho = 0.38$ to 0.51) and two inverted ($\rho = -0.59$ and -0.65; Figure S4). Adjusted Rand index between cluster labels and cancer type was low in both cohorts (0.14 in TCGA, 0.07

in CPTAC), confirming that the clusters capture morphology beyond tissue of origin.

2.6. The bridge extends to protein, and the information added beyond bulk omics is incremental

CPTAC's matched proteomics allowed the bridge to be tested one layer further out. Across 14,687 measurement–gene pairs with both messenger RNA and protein measured, the two modalities' coefficients correlated at $r = 0.63$ (Figure 4C). Among the 320 pairs significant in both, directional concordance was 98.8% (316/320, 95% CI $[96.8\%, 99.7\%]$), and the median protein-minus-RNA difference in effect size was essentially zero ($\Delta\rho = -0.007$). Protein was therefore not globally stronger than RNA, but it was specifically stronger for markers where post-translational biology dominates: nuclear pleomorphism tracked ESR1 protein more closely than *ESR1* transcript in breast carcinoma ($\rho = -0.440$ versus -0.389), consistent with estrogen receptor protein being the phenotypic determinant of luminal differentiation [36], and stromal measurements tracked vimentin protein more closely than transcript ($\rho = 0.578$ versus 0.344). Mitotic index tracked cell-cycle proteins in lung adenocarcinoma (MKI67, CDK1, TOP2A, CHEK1, PLK1), and intratumoral lymphocyte density tracked CD79A, CD3E, GZMA, PRF1, GZMB and IDO1 protein in breast carcinoma. Of the four biological axes probed by this triangulation, proliferation markers are supported in all four lines of evidence, immune and hormonal markers in three or four, and epithelial–mesenchymal markers mainly in TCGA transcripts and CPTAC protein (Figure 4D).

Because protein abundance is largely set by transcript abundance, we asked how much of this signal is protein-specific. Partial Spearman correlations

between each measurement and each protein, adjusting for the matched transcript, left 132 of 14,687 triplets significant after correction (0.9%), 120 of them in breast carcinoma, including stromal and DNA-repair markers such as VIM (Figure 5C); 77% of protein correlations retained more than half their magnitude after adjustment. Immune proteins such as CD8A and GZMB were largely absorbed.

The same question can be asked of the survival results, and gives a correspondingly measured answer. For each of three measurements we refitted the strongest clinical Cox model with matched molecular covariates added (*CD8A*, *TIGIT*, immune subtype and leukocyte fraction for the immune measurements; the G2M_CHECKPOINT score and *MKI67* for mitotic index; Figure 5A). The hazard ratio for intratumoral lymphocyte density in head-and-neck carcinoma was essentially unchanged (0.76 to 0.72, still significant), whereas the hazard ratios for tumor–lymphocyte distance in cervical carcinoma (1.34 to 1.18) and mitotic index in mesothelioma (2.00 to 1.40) moved toward 1 and were no longer significant, so part of those associations is carried by the matched transcripts. Adding a measurement to a clinical-plus-RNA model raised Harrell's apparent (in-sample) concordance index by a median of +0.0007 to +0.0027 across the 18 evaluable cancer types for the three pan-cancer measurements (+0.0038 for nuclear pleomorphism, tested in breast carcinoma only), and the increment was positive in only 50% to 61% of individual cancer types; in breast carcinoma, the largest cohort, intratumoral lymphocyte density added +0.008 over clinical covariates plus *CD8A* (Figure 5B).

2.7. Calibration and sensitivity analyses mark where the atlas is reliable

Because HistoAtlas is meant to be reused, we characterized where its outputs are stable and where they are not. Under a permutation null in which molecular labels were shuffled within cancer type and the full production correction pipeline was re-run (100 permutations over the 39,900 measurement × pathway × cancer-type tests), the mean null discovery rate was 1×10^{-6} against an observed rate of 26.7% (empirical $P = 0.0099$, the smallest attainable with 100 permutations), so the discovery rate is not a trivial consequence of sample size. Replacing Benjamini–Hochberg with the more conservative Benjamini–Yekutieli procedure, which permits arbitrary dependence among tests, retained 68.1% of significant correlations and 54.0% of significant survival associations under a single pooled correction family (Table S11).

Targeted sensitivity analyses locate the confounders that matter. Adjusting for molecular subtype attenuated the leading histomic–molecular correlations by 4.3% in median $|\rho|$ in breast carcinoma and 6.2% in endometrial carcinoma, but by 28% in pancreatic adenocarcinoma, where immune subtype carries much more of the signal (Table S12). Adjusting survival models for leukocyte fraction did not erase the leading prognostic associations; slightly more measurements reached significance afterwards (28/179 versus 25/179), indicating that they are not simply surrogates for immune content (Table S13). Batch effects from tissue source site were present but not dominant: pan-cancer variance decomposition attributed 44.7% of variance to source site, 32.7% to cancer type and 22.6% to residual, while within the 20 cancer types with several sites, source site accounted for a median of 21.3%,

and all per-cancer silhouette scores by source site were negative (Figure S3A). Statistical power is the binding constraint in small cohorts: breast carcinoma can detect HR ≥ 1.62 at 80% power, whereas cholangiocarcinoma ($n = 36$) requires HR ≥ 3.75, which is why 4,846 of 5,623 survival associations carry an *insufficient* evidence badge (Figure S3B).

Some measurements are more protocol-sensitive than others. Median tumor nuclear irregularity shifted systematically between cohorts in all five matched cancer types (standardized difference 1.8 to 2.4 TCGA standard deviations, largest in lung adenocarcinoma; Figure 4A), a scale of shift that would distort any cross-cohort comparison of that measurement. The atlas flags it accordingly. Taken together, these analyses argue that HistoAtlas should be consumed as a calibrated resource rather than an undifferentiated catalog: the measurement layer is portable, the continuous programs are more reliable cross-cohort units than the hard clusters, and the strongest claims are those supported simultaneously by TCGA discovery, CPTAC replication and sensitivity analyses that bound the plausible confounders.

2.8. An open atlas makes pan-cancer morphology reusable by people and by machines

The statistical layer described above is released, not summarized. HistoAtlas is served as a static web application at histoatlas.com, backed by precomputed JSON so that no analysis server is required at runtime, and as a machine-queryable interface exposing 45 typed tools over the same tables. A user can enter from any of four directions: a measurement (what does intratumoral lymphocyte density predict, and where), a cancer type (which measurements matter here), a molecular entity (which morphologies

track *ESR1*), or a slide (what does this case look like relative to the cohort). Each association is displayed with its effect size, confidence interval, adjusted P value, evidence badge, minimum detectable effect, correction-family size and proportional-hazards status, so that a weak result is legible as weak rather than merely present. Measurement tables, per-slide values, cluster assignments and the full association tables are downloadable.

Selecting a survival association leads to the measurement's definition (Figure 6), from there to the per-slide compartment map on which it was computed, and from there to the individual image tiles that contributed most to the value, overlaid with the cell-level predictions that generated them. The contrast between intratumoral and stromal lymphocyte density in breast carcinoma is, at the top of the stack, two hazard ratios (Figure 6B); at the bottom it is the tiles that drive each score on one slide (Figure 6C,D). The same view also exposes what a number hides: the third-ranked intratumoral tile on this slide is dominated by fibroblasts rather than lymphocytes, a case a user can see, question and report. A user who distrusts a number can look at the cells behind it, which is not possible for a resource built on learned embeddings. The programmatic interface exposes the same objects as structured tool calls, so the atlas can also be queried by automated agents conducting systematic hypothesis generation rather than only by humans browsing pages.

3. Discussion

HistoAtlas converts routine H&E pathology into 38 named, compartment-resolved measurements, attaches a fully declared statistical layer to them,

validates that layer in an independent consortium, and publishes the result as queryable infrastructure with provenance back to individual cells. The individual biological correspondences reported here are, for the most part, expected.

What distinguishes this from another predictive computational-pathology system is what the coordinates mean. Foundation-model embeddings are information-dense inputs for predictive models [23, 24], but they are not interpretable: their dimensions have no units, cannot be compared between studies, and do not decompose into quantities a pathologist can check, such as lymphocyte density inside tumor epithelium or distance from a tumor cell to its nearest lymphocyte. Because HistoAtlas measurements are named, they can be given confidence intervals that mean something, corrected within families that can be audited, compared across cohorts on their own scale, and traced back to the tissue that produced them. The cost is expressive power: interpretable measurements will not match learned representations on pure predictive benchmarks.

In the independent CPTAC cohort, the measurement distributions, the correlations between measurements and the correlations with gene expression were reproduced, two of six program distributions were unchanged, and cluster-level pathway enrichment was not reproduced. At the measurement level, four of five matched cancer types passed the prespecified concordance criterion and the inter-feature correlation geometry was preserved, so the coordinates mean approximately the same thing on tissue processed elsewhere. At the biological level, the strongest external evidence came from molecular associations rather than from survival: all ten prespecified histomic–transcriptomic associations

replicated in direction and nine in significance, effect sizes were concordant across 750 matched triples, and the bridge extended to protein abundance with 98.8% directional agreement among doubly-significant pairs. Continuous programs transferred better than discrete clusters, which is a useful negative result: it says that the reusable units of this atlas are the measurements and their sparse combinations, not any particular partition of feature space. Users should treat the ten-cluster solution as a browsing convenience.

Several findings show what the layer is good for. The separation between intratumoral and stromal lymphocyte density quantifies a compartment effect that bulk tumor-infiltrating-lymphocyte scoring collapses [11, 8]; tumor–lymphocyte distance captures immune exclusion as a geometric quantity rather than an impression; mitotic index tracks cell-cycle transcripts and proteins; nuclear pleomorphism tracks luminal differentiation markers, more closely at the protein level than at the transcript level, as estrogen-receptor biology predicts. None of these is conceptually new. The advance is that they are available systematically, pan-cancer, with units and error bars, for anyone to query.

The proteomics analyses position the resource within the wider ecosystem. TCGA anchored pan-cancer genomics and transcriptomics; CPTAC extended it to proteomics [6]; HistoAtlas adds morphology alongside them. The finding that histomic measurements correlate coherently with protein abundance in independent tissue argues that H&E morphology is not merely a low-fidelity shadow of RNA. But the stricter test constrains that claim sharply: after adjustment for matched transcripts only 132 of 14,687 feature–protein triplets remained significant, concentrated in stromal and DNA-repair markers.

HistoAtlas is therefore not an alternative to transcriptomics or proteomics. It is a widely available, spatially resolved, interpretable readout that tracks many molecular states and adds a narrow increment beyond them. The survival analyses point the same way: adding a measurement to a clinical-plus-RNA model raised the apparent concordance index by a median of +0.0007 to +0.0038, with gains in only about half of individual cancer types, and two of three prognostic associations weakened toward the null once matched transcripts were added.

For researchers, the practical value is that a question requiring morphology can now be asked without reprocessing images. A user can ask which measurements associate with a given pathway in a given cancer type, compare a candidate feature's behavior across 21 tumor types, check whether an association survives subtype adjustment, or retrieve the minimum effect a given cohort could have detected before concluding that a null result is informative. The atlas is also available as a Model Context Protocol (MCP) server, which agentic systems can query directly.

Several limitations bound what can be concluded. All associations are observational and cross-sectional; none supports a causal claim. External validation was substantially stronger for molecular associations than for survival, because CPTAC follow-up was absent or minimal in the highest-priority validation cohorts, so clinical-outcome claims rest primarily on TCGA. The measurements inherit the errors of upstream segmentation: the cell model was trained on eight cancer types, and 13 of the 21 included types were processed out of distribution, with reduced detection sensitivity inflating kinetic indices in some of them. Median tumor nuclear irregularity shifted by 1.8 to 2.4 stan-

dard deviations between cohorts and should not be compared across protocols. Molecular subtype adjustment attenuated correlations by 28% in pancreatic adenocarcinoma against under 7% in breast and endometrial carcinoma, so subtype-aware reporting is needed in tumors with strong molecular substructure. Cohort sizes are severely unequal, and 86% of survival associations carry an insufficient-evidence badge, reflecting power rather than absence of signal. Molecular annotation is incomplete; analyzing one representative slide per patient discards intra-patient heterogeneity, although the 312 TCGA patients with several slides make it estimable in future releases; the nine-class taxonomy underlying the measurements does not resolve macrophage or dendritic-cell populations that matter for microenvironment biology, and the atlas has not been benchmarked against foundation-model embeddings or against hybrid representations that combine both.

Those limitations define the next steps: additional external cohorts to map which measurements are protocol-robust, spatial transcriptomics as a stronger ground truth for compartment-resolved immune and stromal quantities, prospective and treatment-annotated cohorts to test clinical utility properly, and a direct benchmark of interpretable measurements against learned embeddings to quantify what interpretability costs. HistoAtlas as it stands is a reference layer: a standardized, externally validated, openly queryable account of what routine cancer pathology contains, built so that other people's questions can be asked of it.

4. Methods

4.1. Cohort construction

Discovery cohort (TCGA).. Formalin-fixed, paraffin-embedded H&E-stained diagnostic whole-slide images were obtained from The Cancer Genome Atlas via the Genomic Data Commons portal for 21 solid-tumor types (Table S1). Slides were excluded if viable tissue area fell below 1 mm^2, if severe processing artifacts were present (pen marks covering >20% of tissue area, or out-of-focus regions), or if essential clinical metadata (vital status, follow-up time) were missing, leaving 6,745 diagnostic slides from 6,088 patients (312 patients contributed more than one slide). Slide-level descriptive analyses (the inter-measurement correlation geometry, the UMAP embedding and pan-cancer clustering) used all 6,745 slides. To avoid pseudo-replication, every statistical association used one representative slide per patient, chosen by a deterministic priority ranking (quality-control pass status, then tissue-area adequacy, then tile-extraction coverage, then tile count, then slide name). Twelve further TCGA cancer types were excluded in their entirety because their dominant cell morphologies fall outside the training domain of the cell detection model (Table S2). GDC file UUIDs for all 6,745 slides are listed in the code repository.

Matched clinical data (overall, disease-specific, disease-free and progression-free survival; age at diagnosis, sex, pathologic stage, tissue source site and tumor purity) came from the TCGA Pan-Cancer Clinical Data Resource [37]. Molecular data comprised RNA-seq expression (RSEM-normalized), somatic mutations from the MC3 multi-caller ensemble [38], copy-number variation, immune cell fractions from CIBERSORT [39] and xCell [40], tumor purity from ABSOLUTE [41], and immune subtypes (C1–C6) from Thorsson et al. [42].

Molecular data were matched to slides by TCGA case barcode (first 12 characters); for patients with multiple aliquots the primary-tumor aliquot was used. Of 6,745 slides, 5,590 (82.9%) had matched immune subtype labels.

Validation cohort (CPTAC).. Diagnostic H&E slides and matched transcriptomic, proteomic and clinical data were obtained for five CPTAC cancer types with TCGA counterparts: breast carcinoma (198 slides), lung adenocarcinoma (240), head-and-neck squamous carcinoma (390 slides from 112 patients), pancreatic adenocarcinoma (175) and endometrial carcinoma (92), totalling 1,095 slides from 817 patients. The identical pipeline, model weights and feature definitions used for TCGA were applied without retraining, recalibration or cohort-specific tuning. For patients with multiple slides, one representative slide was selected by a deterministic priority ranking (quality-control pass status, then tissue-area adequacy, then tile-extraction coverage, then tile count, then fewest missing measurements, then slide name), and all downstream analyses were performed at patient level. Global tandem-mass-tag proteomics were available for four of the five types (head-and-neck squamous carcinoma is a GDC-only study without a proteomic profile).

4.2. Tissue segmentation

Tissue segmentation used a CellViT-inspired architecture [43] with a Phikon self-supervised ViT-B backbone [26], trained on the PanopTILs crowdsourced annotation dataset [44]. Inference was performed at 0.5 µm/px on 224×224 pixel tiles with 32-pixel overlap, with the final mask obtained by majority voting in overlap regions. The model assigns nine tissue classes (cancerous epithelium, stroma, necrosis, normal epithelium, tumor-infiltrating

lymphocytes, junk/debris, blood, other, whitespace); mean intersection-over-union was 0.72 on the PanopTILs held-out test set, with substantial per-class variation and near-zero recall for necrosis and normal epithelium in deployment, reflecting their rarity in both the training set and resected TCGA specimens. Following the International Immuno-Oncology Biomarker Working Group recommendation [33], regions classified as tumor-infiltrating lymphocytes were reassigned to stroma before all downstream computation, yielding five effective compartments: cancerous epithelium, stroma, necrosis, normal epithelium and blood.

4.3. Cell segmentation and classification

Cell instance segmentation and classification used HistoPLUS [32], which classifies cells into 13 types; the nine used here are tumor cells, lymphocytes, fibroblasts, plasmocytes, neutrophils, eosinophils, red blood cells, apoptotic bodies and mitotic figures (mean panoptic quality 0.509; per-class 0.28 to 0.73, lowest for rare types). Inference used 224×224 pixel tiles at $40\times$ magnification (0.25 µm/px) when available, falling back to $20\times$ (0.50 µm/px); most slides were scanned at $40\times$. Tiles were extracted with a 64-pixel (16 µm) overlap margin and cells in overlap regions were deduplicated by a union-find procedure merging instances whose centroids fall within 10 µm, roughly the diameter of an epithelial nucleus. The cell model was trained on pathologist annotations from eight cancer types (LUAD, LUSC, BRCA, COAD, BLCA, OV, PAAD, MESO); the remaining 13 included types were processed out of distribution. Tissue–cell discordance (cells inside a tissue-model tumor region not classified as tumor cells) ranged from 0% to 12% across cancer types, highest in pancreatic adenocarcinoma (12.2%) despite being in distribution,

plausibly because of dense desmoplastic stroma and small tumor glands; out-of-distribution types additionally showed inflated kinetic indices in prostate, liver and thyroid carcinoma attributable to reduced detection sensitivity. Red blood cells and plasmocytes were not used in measurement computation.

4.4. Spatial compartment definition

All spatial computations were performed on compartment masks resampled to a common 8 µm/px grid by nearest-neighbor interpolation, so that boundary quantities are scanner-invariant. Connected components below $A_{\min} = 2{,}048$ µm^2 were removed per compartment before distance-transform computation, to prevent segmentation fragments from inflating boundary lengths. Five spatial bands were defined from the signed Euclidean distance transform d_T to the tumor boundary (the outer contour of the cancerous epithelium mask; positive inside tumor): tumor front ($0 \leq d_T \leq 50$ µm), tumor core ($d_T > 50$ µm), near stroma ($-50 \leq d_T < 0$), far stroma ($-200 \leq d_T < -50$) and necrosis ring (within 100 µm of the necrosis boundary). The 50 µm front band corresponds to approximately five cell diameters; the 200 µm stromal cutoff is a design parameter of the pipeline. Slides were assigned a growth-pattern regime from the tumor front fraction $\varphi = A(B_T^{0\text{-}50})/A(\Omega_T)$: mass-forming ($\varphi \leq 0.5$), intermediate ($0.5 < \varphi \leq 0.8$) or infiltrative ($\varphi > 0.8$). Distance-based measurements are quantized at 8 µm owing to the grid spacing.

4.5. Histomic measurement definitions

We computed 38 quantitative histomic measurements per slide. For presentation they are grouped into six categories: tissue composition (3), cell densities (8), nuclear morphology and kinetics (7), spatial organization (11),

spatial heterogeneity (5) and cell-type ratios (4). The atlas's machine-readable feature metadata uses a coarser five-way grouping of the same 38 measurements (composition 3, density 6, morphology 8, spatial 18, heterogeneity 3); the two schemes partition the identical set and no measurement differs between them. Complete definitions, units and both category assignments are given in Table S3.

Two candidate measurements (normal epithelium area fraction and tumor–normal contact fraction) were removed from the atlas entirely because the tissue model did not reliably detect normal epithelium and both showed zero signal across all slides. Seven ratio measurements were susceptible to unstable values when denominators approach zero; these were controlled by winsorization (§4.6).

4.6. Measurement preprocessing

Preprocessing proceeded in three steps. First, 22 measurements with heavy right-skew (all densities, ratios, distances and heterogeneity measures) were log-transformed as $\log(1+x)$. Second, all 38 were winsorized at the 0.5th and 99.5th percentiles computed per measurement across the full cohort; global rather than per-cancer percentiles were used because the target artifacts (segmentation failures, ratio instabilities) occur across cancer types and because several cohorts are too small (for example cholangiocarcinoma, $n = 38$) for reliable percentile estimation. Third, measurements were z-score standardized. The scope of standardization varied by analysis: UMAP and K-means used global z-scores; Cox regression used per-cancer-type z-scores, so hazard ratios are per standard deviation within cancer type; Spearman correlations used no standardization, being rank-based. Missing values (<2%

of entries) were imputed with the column median before standardization. Raw values are retained for display in the web interface; all models consumed preprocessed values.

4.7. Survival analysis

Associations between each measurement and four endpoints (overall survival, designated primary; disease-specific, disease-free and progression-free survival) were assessed by Cox proportional hazards regression using `lifelines` [45] without penalization. Each model tests a single measurement plus covariates; no multivariate models combining several measurements are included in the atlas. Two specifications were fitted per measurement, cancer type and endpoint: unadjusted (measurement only) and adjusted (measurement, age at diagnosis, sex and pathologic stage, stratified by tissue source site). Stratification lets each site carry its own baseline hazard without consuming degrees of freedom, the standard approach for multi-center data [46]. Pan-cancer models additionally stratified on cancer type, estimating one shared coefficient across per-type baseline hazards. Hazard ratios are reported per one standard deviation of the preprocessed measurement, with 95% Wald confidence intervals from the observed Fisher information.

For adjusted models, missing covariates were handled by multiple imputation by chained equations (`scikit-learn IterativeImputer` with Bayesian ridge, 5 imputations, maximum 10 iterations, seed incremented per imputation), pooled by Rubin's rules, with the fraction of missing information recorded. When covariate missingness exceeded 20% in any variable, total sample size fell below 30, or fewer than 3 of 5 imputations converged, complete-case analysis was used instead. Missingness arose mainly from incomplete

staging (Table S6). A minimum of $n \geq 30$ with $\geq$10 events was required, and models were fitted only when residual degrees of freedom remained positive.

The proportional-hazards assumption was tested for every fitted model using Schoenfeld residuals with the Kaplan–Meier time transform [47], flagged as pass ($P \geq 0.05$), warn ($0.01 \leq P < 0.05$) or fail ($P < 0.01$). When the assumption was violated ($P < 0.05$, covering both warn and fail), the hazard ratio, confidence interval and P value were invalidated, and the P value was set to 1.0 for correction so that family size is preserved without inflating discoveries. Of 5,623 fitted models, 5,283 yielded valid estimates. Restricted mean survival time (RMST) differences were computed as an assumption-free complement, as the area under the Kaplan–Meier curve to a cancer-type-specific horizon (default 1,095 days; 2 years for pancreatic adenocarcinoma and mesothelioma; 5 years for thyroid and prostate carcinoma; Table S7) with Irwin variance [48], tested by 5,000-permutation inference on median-split groups with 1,000-resample percentile bootstrap intervals, requiring $n \geq 20$ with $\geq$5 per group. We note that dichotomization for RMST discards information relative to continuous methods [49]. Cluster-level survival used two-sided log-rank tests of members against non-members.

4.8. Molecular correlations

Spearman rank correlations were computed between each of the 38 measurements and 322 molecular targets: 133 curated genes drawn from OncoKB [50] and the COSMIC Cancer Gene Census [51], immune checkpoint targets and EMT/stemness markers [52] (Table S4), each assessed for expression and copy number (133×2); the 50 MSigDB Hallmark pathway scores [31]; and 6 CIBERSORT immune cell fraction scores. Pathway activity was computed

by single-sample gene set enrichment analysis (`gseapy` 1.1.12) [53, 54] on the batch-corrected TCGA Pan-Cancer expression matrix from UCSC Xena [55], excluding gene sets with fewer than 10 genes present after intersection.

Unadjusted models used `scipy.stats.spearmanr` with analytical t-approximation P values, chosen over permutation to avoid P-value banding, with 95% percentile bootstrap intervals from 1,000 resamples. Covariate-adjusted models used partial Spearman correlations computed by rank-transforming measurement, target and covariates, residualizing the ranked measurement and target against ranked covariates by ordinary least squares, and correlating the residuals [56]; P values used the standard partial-correlation t statistic with $\mathrm{df} = n - 2 - k$ for k one-hot-encoded covariate columns. Tissue source site entered as a grouped covariate (five most frequent sites per cancer type, remainder collapsed to "Other") rather than a stratification variable, since rank residualization requires explicit columns. A cancer type was analyzed if $n \geq 30$; each pair required $n \geq 10$ non-missing observations. Per-task random seeds were derived deterministically from a hash of the cancer type and feature names. The analysis comprised 487,638 evaluable pairs (measurements $\times$ 322 targets $\times$ 22 cohorts $\times$ 2 models, excluding combinations with insufficient data). Of these, 461,130 involve the 38 measurements; the remaining 26,508 involve three slide-level quality-control variables (missingness flag, number of missing measurements and number of outlying measurements) that were computed alongside the measurements, are released in the same table, and entered the same correction families.

4.9. Categorical associations

Associations with categorical molecular variables (somatic mutation status, copy-number alteration, immune subtype) were tested per cancer type. Unadjusted two-group comparisons used the two-sided Mann–Whitney U test with Cliff's δ as effect size, computed as the mean of $\mathrm{sign}(x_i - y_j)$ over all pairs with 1,000-resample bootstrap intervals (bootstrap rather than analytical intervals because ties violate the continuity assumption of the analytical variance formula [57]). Variables with more than two levels used the Kruskal–Wallis H test with $\eta^2_H = (H - k + 1)/(N - k)$ [58], floored at zero. Covariate-adjusted models used rank-ANCOVA with Freedman–Lane permutation inference [59] (1,000 permutations of covariate-only residuals), with partial η^2 as effect size. Group ordering followed a fixed convention (mutant before wild-type, altered before neutral) so that the sign of Cliff's δ is interpretable across analyses. A minimum of $n \geq 30$ total and ≥ 5 per group was required.

4.10. Clustering and embedding

Unsupervised clustering was performed at two levels. At the pan-cancer (L1) level, all 6,745 slides were clustered on the full 38-dimensional preprocessed vector; at the cancer-specific (L2) level, clustering was performed within each cancer type using cancer-type-specific z-scores. We used K-means (`scikit-learn`, $n_{\mathrm{init}} = 10$, random state 42). For L1, solutions were computed for $K \in \{3, \ldots, 25\}$ and scored by silhouette, Calinski–Harabasz, Davies–Bouldin and gap statistics; $K = 10$ was selected on convergence of silhouette (local maximum), Calinski–Harabasz (plateau) and Davies–Bouldin (local minimum), the gap statistic favouring $K = 8$. No dimensionality

reduction was applied before clustering, to keep cluster feature profiles interpretable; the effective dimensionality of 18 components at 90% explained variance implies moderate redundancy, which gives correlated measurements proportionally more weight. For L2, K was selected by the elbow method over $K \in [2, 8]$ [60], yielding 69 subclusters; cancer types with fewer than 20 samples received a single cluster and individual clusters with fewer than 10 samples were dropped. Stability was assessed by 50 iterations of 80% subsampling without replacement, re-clustering and comparing to the original labels by adjusted Rand index and mean best-match Jaccard index [61]. Cluster names were generated from a fixed schema combining an immune axis label, a stromal axis label, the most extreme remaining measurement, and a cancer-enrichment tag where one type constitutes $\geq$40% of the cluster. Two-dimensional visualization used UMAP ($n_{\text{neighbors}} = 15$, min_dist $= 0.1$, Euclidean, random state 42) [62] for display only; no inference was drawn from embedding coordinates and all clustering was performed in the original 38-dimensional space.

4.11. Continuous histomic programs

Sparse principal component analysis (`scikit-learn SparsePCA`, 6 components, $\alpha = 1.0$, random state 42) was fitted on standardized measurements from 5,371 TCGA patients with complete feature vectors, after representative-slide selection. Programs were named automatically from their top-three loading measurements and interpreted post hoc by Spearman correlation against all 50 Hallmark pathway scores in TCGA. The fitted scaler and decomposition were then applied unchanged to 297 CPTAC patients with complete feature vectors. Distributional transfer was assessed by two-sample Kolmogorov–

Smirnov tests between TCGA and CPTAC program scores. Program–protein associations were computed as Spearman correlations against CPTAC tandem-mass-tag protein abundance (2,220 program–protein pairs across four cancer types).

4.12. External validation in CPTAC

Measurement concordance.. For each matched cancer-type pair we computed the standardized difference $d = (\text{median}_{\text{CPTAC}} - \text{median}_{\text{TCGA}})/\text{SD}_{\text{TCGA}}$ per measurement over all slides of that type in each cohort (Figure 4A shows the signed value). A measurement passes at $|d| \leq 1.0$; a cancer type passes when more than 80% of measurements ($\geq 31/38$) pass. The inter-measurement correlation geometry was compared by computing the full 38×38 Spearman matrix separately in each cohort and correlating the flattened upper triangles (703 pairs) by Pearson correlation, with a prespecified pass criterion of $r > 0.80$.

Association replication.. Ten histomic–molecular associations were prespecified from the TCGA analysis before CPTAC results were examined. Their CPTAC P values were corrected within the full CPTAC correction family (approximately 24,700 expression tests per model per cancer type), not within the prespecified subset, so replication is assessed against the same multiplicity burden as discovery. Direction concordance and significance replication are reported with 95% Clopper–Pearson exact binomial intervals. Aggregate transfer was quantified by extracting the top 50 correlated genes per measurement per cohort for 8 representative measurements across 4 matched cancer types, matching by gene symbol, and correlating the two cohorts' Spearman

coefficients across the resulting 750 triples by Pearson correlation with Fisher z-transform intervals.

Survival replication.. Eight prespecified survival hypotheses were tested. Following ICH E9 guidance for confirmatory tests of prespecified hypotheses, nominal (uncorrected) P values are reported and no false-discovery correction was applied. Direction concordance is computed only over associations that were significant in TCGA and for which CPTAC had sufficient follow-up. Survival follow-up was absent or minimal in breast (0 patients with overall survival data), lung adenocarcinoma (25) and endometrial carcinoma (8), leaving pancreatic adenocarcinoma ($n = 110$) and head-and-neck squamous carcinoma ($n = 41$) as the only evaluable cohorts; their minimum detectable effects at 80% power are HR $\geq$ 1.71 and HR $\geq$ 2.45 respectively, so this analysis is reported as underpowered by construction.

Cluster-structure transfer.. A `StandardScaler` and K-means model ($K = 10$) fitted on TCGA were applied unchanged to CPTAC by nearest-centroid prediction. Per-cluster pathway enrichment (Cliff's δ from Mann–Whitney U tests, in-cluster versus out-of-cluster) was computed independently in each cohort and compared by Spearman correlation, both per cluster and across all 450 matched cluster–pathway pairs. Adjusted Rand index, normalized mutual information and V-measure between cluster labels and cancer type were computed in both cohorts.

4.13. Proteomics and protein-specific signal

Spearman correlations were computed between each measurement and CPTAC global tandem-mass-tag protein abundance in the four cancer types

with proteomic profiles. Direction concordance between messenger RNA and protein was assessed over pairs significant in both modalities after correction, with Clopper–Pearson intervals. To determine how much of the protein signal is independent of transcript abundance, we computed partial Spearman correlations between each measurement and each protein adjusting for the matched transcript, using the same rank-residualization procedure as §4.8, over 14,687 (measurement, gene, cancer type) triplets, with Benjamini–Hochberg correction applied per cancer type. Retention was summarized as the fraction of triplets for which the partial coefficient retained more than half the magnitude of the unadjusted coefficient.

4.14. Added-value analyses

To test whether measurements carry information beyond bulk transcriptomics, Cox models for four target measurements were refitted with matched molecular covariates added to the clinical specification: *CD8A*, *TIGIT*, immune subtype and leukocyte fraction for intratumoral lymphocyte density and tumor–lymphocyte nearest-neighbor distance; HALLMARK_G2M_CHECKPOINT and *MKI67* for mitotic index; and *ESR1* plus PAM50 subtype for nuclear pleomorphism in breast carcinoma. Incremental discrimination was quantified as Harrell's concordance index for four nested models per cancer type (clinical only; clinical + RNA; clinical + histomics; clinical + RNA + histomics), summarized as the median across the 18 cancer types with evaluable models, together with the fraction of cancer types in which the increment was positive. Concordance was evaluated on the data used for fitting, so the increments are apparent rather than cross-validated and are an upper bound on out-of-sample gain.

4.15. Multiple testing correction

All P values were corrected by the Benjamini–Hochberg procedure [63] within explicitly declared families, chosen to pool biologically coherent tests without the conservatism of pooling unrelated analyses (Table S5 gives every family definition). Survival families were defined per cancer type × endpoint × model, so that correction pools only the 38 measurement-level tests within one biological context; correlation families were defined per cancer type × molecular data type × model, giving 150 families. Every adjusted P value is stored alongside its correction-family identifier and family size, so that any reported result can be re-derived. Gene set enrichment analysis used its own canonical false-discovery procedure based on pooled null normalized enrichment score distributions [64] rather than Benjamini–Hochberg. Overall survival is the primary endpoint; the other three are sensitivity analyses, corrected independently per endpoint rather than pooled, because the endpoints share events and therefore violate the independence assumption of joint correction. Permutation P values used the add-one correction of Phipson and Smyth [65]. Bootstrap intervals used 1,000 percentile resamples and were reported only when ≥90% of resamples produced valid estimates. Unless stated otherwise, "significant" means $P_{\text{adj}} < 0.05$.

4.16. Power analysis and evidence grading

The minimum detectable effect size at 80% power ($\alpha = 0.05$, two-sided) was computed for every association: by the Schoenfeld–Freedman approximation [66, 67] for survival, the Fisher z-transform for correlations, and simulation-based power curves (2,000 simulated datasets with binary search) for categorical associations. Every association carries an evidence badge:

strong ($P_{\text{adj}} < 0.01$, effect above the medium-effect threshold, narrow interval, $n \geq 100$), **moderate** ($P_{\text{adj}} < 0.05$, effect above the small-effect threshold, narrow or moderate interval, $n \geq 50$), **suggestive** ($P_{\text{adj}} < 0.10$ or interval excluding the null, $n \geq 30$) or **insufficient** ($n < 30$ or missing statistics). Thresholds for ρ and η^2 follow Cohen [68], those for Cliff's δ approximate Romano et al. [69], and those for hazard ratios were set for this study: medium, HR ≥ 1.5 or ≤ 0.667, $|\rho| \geq 0.3$, $|\delta| \geq 0.3$, $\eta^2 \geq 0.06$; small, HR ≥ 1.18 or ≤ 0.847, $|\rho| \geq 0.1$, $|\delta| \geq 0.15$, $\eta^2 \geq 0.01$.

4.17. Quality control, batch effects and sensitivity analyses

Batch effects from tissue source site were assessed by principal variance component analysis [70], decomposing variance in the top principal components (explaining ≥80% cumulative variance, up to 10 components) into site, cancer type and residual fractions using marginal one-way ANOVA η^2 weighted by explained variance. Because site and cancer type are not orthogonal, marginal η^2 values can exceed 1.0 in sum; residual variance was computed as $\max(0, 1 - \eta^2_{\text{batch}} - \eta^2_{\text{bio}})$ and all three fractions renormalized. Silhouette analysis treated site labels as cluster assignments on standardized measurements (Euclidean, subsampled to 5,000 slides), with scores above 0.25 flagged as moderate and above 0.5 as strong batch effects.

Four sensitivity analyses were run. (i) A permutation null shuffled molecular labels within cancer type and re-ran the full production correction pipeline over the 39,900 measurement × pathway × cancer-type tests for 100 permutations. (ii) The Benjamini–Yekutieli procedure, valid under arbitrary dependence, was substituted for Benjamini–Hochberg; for this comparison all tests of a given analysis type were pooled into a single family, so its

denominator (272 significant survival associations) differs from the production 149-family design (260). (iii) Top correlations were recomputed with molecular subtype as an additional covariate (PAM50 for breast, molecular class for endometrial, immune subtype for pancreatic carcinoma; the lung adenocarcinoma driver stratification was uninformative because neither *EGFR* nor *KRAS* mutation was prevalent enough to form contrasting groups). (iv) Survival models for the top 10 measurements were refitted with leukocyte fraction as an additional covariate.

4.18. Software, reproducibility and interfaces

All analyses were implemented in Python 3.11 and orchestrated as a directed acyclic graph by Snakemake [71], which supports a subsetted dry-run mode for validating the full pipeline before production runs. Key versions: `lifelines` 0.29.0, `scipy` 1.12.0, `scikit-learn` 1.4.0, `umap-learn` 0.5.5, `gseapy` 1.1.12, `statsmodels` 0.14.1, `numpy` 1.26.4, `pandas` 2.2.0. Every stochastic procedure (permutation, bootstrap, subsampling, K-means initialization, UMAP) used an explicit seed; per-task seeds in embarrassingly parallel stages were derived deterministically from task identifiers so that results do not depend on execution order. Each run writes a provenance record containing pipeline version, timestamps, input checksums, an output manifest and an automated validation report.

The web interface is an Astro static site with React components; all statistical results, feature profiles, cluster metadata and visualization data are serialized to static JSON at build time, so no analysis server runs at request time. Three interpretability layers connect statistics to tissue: a per-slide compartment map rendering the five tissue compartments across

nine spatial zones; for each measurement, the top-ranked 224 × 224 pixel tiles scored by that measurement's own computation strategy, each with slide coordinates, a scale bar and a toggleable cell-type overlay. The overlay renders the full set of classes the cell model emits (14 in the current release) for visual inspection, whereas the nine classes listed in §4.3 are those used to compute the measurements; the remaining classes are displayed but not quantified. The interface also provides bidirectional links from association tables to measurement pages and from measurement pages to specific tiles on specific slides. The same tables are exposed programmatically through 45 typed tools over a Model Context Protocol server, so that automated agents can issue the same queries as human users.

4.19. Figure preparation

Every figure panel was drawn from data exported directly from the frozen pipeline outputs (TCGA run of 9 March 2026; CPTAC run of 17 March 2026) by a single script (`patterns_research_article/scripts/export_figure_data.py`), which writes one machine-readable file per figure containing every plotted value and every statistic quoted in the legends. A second script (`build_figures.py`) renders these files deterministically as vector graphics at print size, and a third (`export_figures.py`) produces the submitted PDF and TIFF files; rerunning the three scripts regenerates every figure from the released tables. Hematoxylin-and-eosin tiles and cell-call overlays in Figures 1A and 6 are the pipeline's own image files, reproduced without alteration other than uniform scaling; compartment maps are label images in which the pipeline's class colors were remapped to the figure palette. Dense scatter plots are shown as two-dimensional histograms with logarithmic color scaling, and point clouds

in Figure 3C are shown on percentile-rank scales, on which a Spearman coefficient is the Pearson coefficient of the plotted points.

Resource availability

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Pierre-Antoine Bannier (pierrean-toine.bannier@gmail.com).

Materials availability

This study did not generate new materials or reagents.

Data and code availability

- This paper analyzes existing, publicly available data. TCGA diagnostic whole-slide images, clinical annotations and molecular profiles were obtained from the Genomic Data Commons (`https://portal.gdc.cancer.gov`); the file names of all 6,745 slides are listed in the derived-data archive below. Pan-cancer clinical endpoints were taken from the TCGA Pan-Cancer Clinical Data Resource [37], somatic mutations from MC3 [38], immune subtypes from Thorsson et al. [42], and batch-corrected expression from UCSC Xena (`https://xenabrowser.net`). CPTAC whole-slide images and transcriptomic, proteomic and clinical data were obtained from the Genomic Data Commons and the Proteomic Data Commons (`https://proteomic.datacommons.cancer.gov`). No new primary patient data were generated.

- All derived data reported in this paper (per-slide measurement tables for both cohorts, the 487,638 molecular correlations, the 5,623 survival models, categorical and treatment associations, cluster assignments and enrichments, program loadings and scores, and all external-validation and sensitivity outputs) are downloadable from https://histoatlas.com (RRID:SCR_028056). The values plotted in every figure panel are provided as machine-readable files in the code repository (`patterns_research_article/figure_data`). The derived tables and figure data files are archived at Zenodo (https://doi.org/10.5281/zenodo.24014110) and are publicly available.

- All original code is publicly available at https://github.com/histoatlas/histoatlas, including the analysis library, the Snakemake workflow, the scripts that regenerate every figure from the released tables and the static site. **TODO(author): archive the commit corresponding to this manuscript and insert its DOI here.**

- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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Author contributions

P.-A.B.: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing – original draft, writing – review and editing, and project administration.

Declaration of interests

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to restructure the manuscript into the Patterns format and edit it for language and readability, to check quantitative statements against the released data tables, and to write the code that draws the figures from those tables and compose them in Claude Design. No figure value, histology image or result was generated or altered by the tool; the figure-preparation procedure is described in Methods. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Supplemental information

Document S1. Figures S1–S4 and Tables S1–S13.

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Figure titles and legends

Figure 1. HistoAtlas converts routine pathology into a named, compartment-resolved measurement layer. (A) Pipeline, illustrated on one lung adenocarcinoma slide (TCGA-MP-A4T9). Diagnostic H&E whole-slide images (6,745 slides from 6,088 patients, 21 TCGA cancer types) are segmented into tissue compartments and spatial bands and into individual cells (H&E tile and cell-call overlay, 224 × 224 pixels), from which 38 measurements in six groups are computed per slide. A statistical layer links each measurement to outcomes and molecular data, and the results are served through a web atlas and a programmatic interface (screenshot of the atlas). This panel is an explanatory schematic composed in Claude Design (Anthropic) from pipeline images that were not altered. (B) Spearman correlation between the 38 measurements across all slides with complete measurements ($n = 5{,}916$), ordered by Ward-linkage clustering (dendrogram above). The diagonal is masked; the strip beside each row gives the measurement group. (C) UMAP embedding of all 6,745 slides colored by cancer type, with kernel-density contours (gray) and labels at the median position of the larger cancer types. The embedding is used for display only. (D) Cancer-type composition of the ten pan-cancer K-means clusters (slides per cluster in parentheses); types making up at least 20% of a cluster are labeled. The right column gives the overall-survival hazard ratio of cluster membership where significant (Cox regression stratified by cancer type, Benjamini–Hochberg adjusted $P < 0.05$; down triangle, lower hazard; up triangle, higher hazard; dash, not significant). (E) Mean z-scored measurement value per cluster (rows ordered as in D), clipped at ± 2; measurements are ordered by Ward-linkage clustering and the

strip above gives the measurement group.

Figure 2. Six continuous histomic programs summarize the measurement space and partially transfer across cohorts. (A) Sparse principal component loadings of the 38 measurements on six programs, fitted on 5,371 TCGA patients with complete measurement vectors ($\alpha = 1.0$). Measurements are grouped by the program on which they load most strongly. Program names summarize the pathway correlations in (C). (B) Distributions of program scores in TCGA (blue, $n = 5{,}371$) and in CPTAC after projection with the unchanged TCGA-fitted transform (orange, $n = 297$). Violins are kernel density estimates; boxes show the interquartile range, the white line the median and whiskers the 5th to 95th percentiles. D and P, two-sample Kolmogorov–Smirnov test. (C) Spearman correlation between each program and each of the 50 MSigDB Hallmark pathway activity scores in TCGA ($n = 5{,}371$), pathways ordered by Ward-linkage clustering. Dots mark Benjamini–Hochberg adjusted $P < 0.05$ across the 300 tests.

Figure 3. Compartment-resolved immune topology is prognostic and molecularly anchored. (A) Hazard ratios for overall survival per standard deviation of intratumoral (blue circles) and stromal (orange diamonds) lymphocyte density, from Cox models adjusted for age, sex and pathologic stage and stratified by tissue source site (pan-cancer models additionally stratified by cancer type). Lines are 95% Wald confidence intervals, truncated at the axis limits. Filled markers denote moderate or strong evidence badges; open markers suggestive or insufficient evidence. Cancer types without a valid adjusted model for either measurement (PRAD, THYM, UCS) are omitted; n, patients per model. (B) Kaplan–Meier estimates of overall survival in breast

carcinoma for patients above and below the median intratumoral lymphocyte density ($n = 480$ each; shaded bands, 95% confidence intervals; ticks, censoring), with numbers at risk below. The hazard ratio shown is from the adjusted Cox model on the continuous measurement, not from the dichotomized groups. (C) Tumor–lymphocyte nearest-neighbor distance at the invasive front against *CD8A* expression in breast carcinoma, both as percentile ranks ($n = 958$ patients). The line is the least-squares fit on ranks, whose slope equals Spearman ρ. Statistics are from the unadjusted model; Spearman correlation with analytical t-approximation P value, Benjamini–Hochberg adjusted within its declared family; 95% bootstrap confidence interval. (D) The ten strongest positive and five strongest negative gene correlates of intratumoral lymphocyte density among significant associations in breast carcinoma ($n = 953$, adjusted for age). Bars are Spearman ρ, error bars 95% bootstrap confidence intervals (1,000 resamples); immune-checkpoint genes are in bold.

Figure 4. The measurement layer and its molecular links transfer to an independent consortium and extend to protein. (A) Standardized difference for each measurement and matched cancer type, computed as (median in CPTAC − median in TCGA) / standard deviation in TCGA over all slides of that type (CPTAC n: BRCA 198, HNSCC 390, LUAD 240, PAAD 175, UCEC 92; TCGA n: 1,037, 471, 511, 146, 459). Dots mark $|d| > 1$; the numbers above each column count measurements with $|d| \leq 1$ (prespecified pass, ≥ 31 of 38). Rows follow the measurement groups (strip, colors as in Figure 1). (B) Spearman coefficients of the same measurement–gene pairs (expression, unadjusted) in TCGA and CPTAC, across all 25,155 pairs evaluable in both cohorts, shown as a two-dimensional histogram (log color

scale); dashed line, identity; r, Pearson correlation across pairs. (C) Messenger RNA versus protein Spearman coefficients for the same measurement and gene in the four CPTAC cancer types with proteomics (n = 14,687 pairs, histogram as in B). Orange points are the 320 pairs significant in both modalities, of which 316 have the same sign (98.8%, 95% Clopper–Pearson interval 96.8% to 99.7%). (D) Evidence triangulation for representative measurement–gene pairs grouped by biological axis. A filled circle denotes significance (false discovery rate below 0.05) in at least one cancer type for TCGA messenger RNA, CPTAC messenger RNA and CPTAC protein; the survival column marks a significant overall-survival association of the measurement itself in TCGA (unadjusted model) and is therefore shared by all genes of an axis. Open circles denote non-significance, which for CPTAC frequently reflects limited power rather than a contradicted effect.

Figure 5. Histomic measurements add a small increment beyond bulk molecular measurements. (A) Overall-survival hazard ratios (95% confidence intervals) before (blue circles; age, sex and, where available, stage) and after (orange triangles) adding matched molecular covariates: *CD8A*, *TIGIT*, immune subtype and leukocyte fraction for the two immune measurements; HALLMARK_G2M_CHECKPOINT and *MKI67* for mitotic index. For each measurement, the cancer type with the strongest clinical-model association is shown; n, patients. (B) Harrell's concordance index of four nested Cox models in breast carcinoma (n = 940, 127 events): clinical covariates only, plus the matched transcript (named at left), plus the measurement, and plus both. Values are apparent (in-sample) concordance. (C) Spearman correlation between each measurement and each protein before (horizontal axis) and

after (vertical axis) partialling out the matched transcript, for 14,686 triplets with both values in four CPTAC cancer types (two-dimensional histogram, log color scale; dashed line, identity). Orange points are the 132 triplets that remain significant after adjustment (Benjamini–Hochberg per cancer type; 120 in breast carcinoma); 77% of triplets retain more than half of their unadjusted magnitude.

Figure 6. From cohort statistics back to cells: spatial provenance in HistoAtlas. (A) Tissue compartment map of a breast carcinoma slide (TCGA-A1-A0SE) showing the spatial bands defined by the signed distance to the tumor boundary; the pipeline's label colors were remapped to the palette shown. Circles mark the three highest-scoring tiles for intratumoral (blue, 1) and stromal (orange, 2) lymphocyte density; the largest circle marks the top tile. (B) The two overall-survival associations that a user reaches from this slide (adjusted Cox models in breast carcinoma, $n = 960$, 135 events; hazard ratio per standard deviation with 95% confidence interval). (C and D) The three highest-scoring 224×224 pixel tiles on this slide for intratumoral (C) and stromal (D) lymphocyte density, as H&E (top) and with the cell calls of the cell model outlined (bottom), with the tile score used to rank them. Images are the atlas's own tiles, shown unaltered. Lymphocyte calls (green outlines) dominate the two top-ranked tiles of each measurement, whereas the third-ranked intratumoral tile contains mostly fibroblast calls (blue), an example of the inspection the provenance path allows. The legend lists the outline colors of the 14 classes rendered by the atlas.

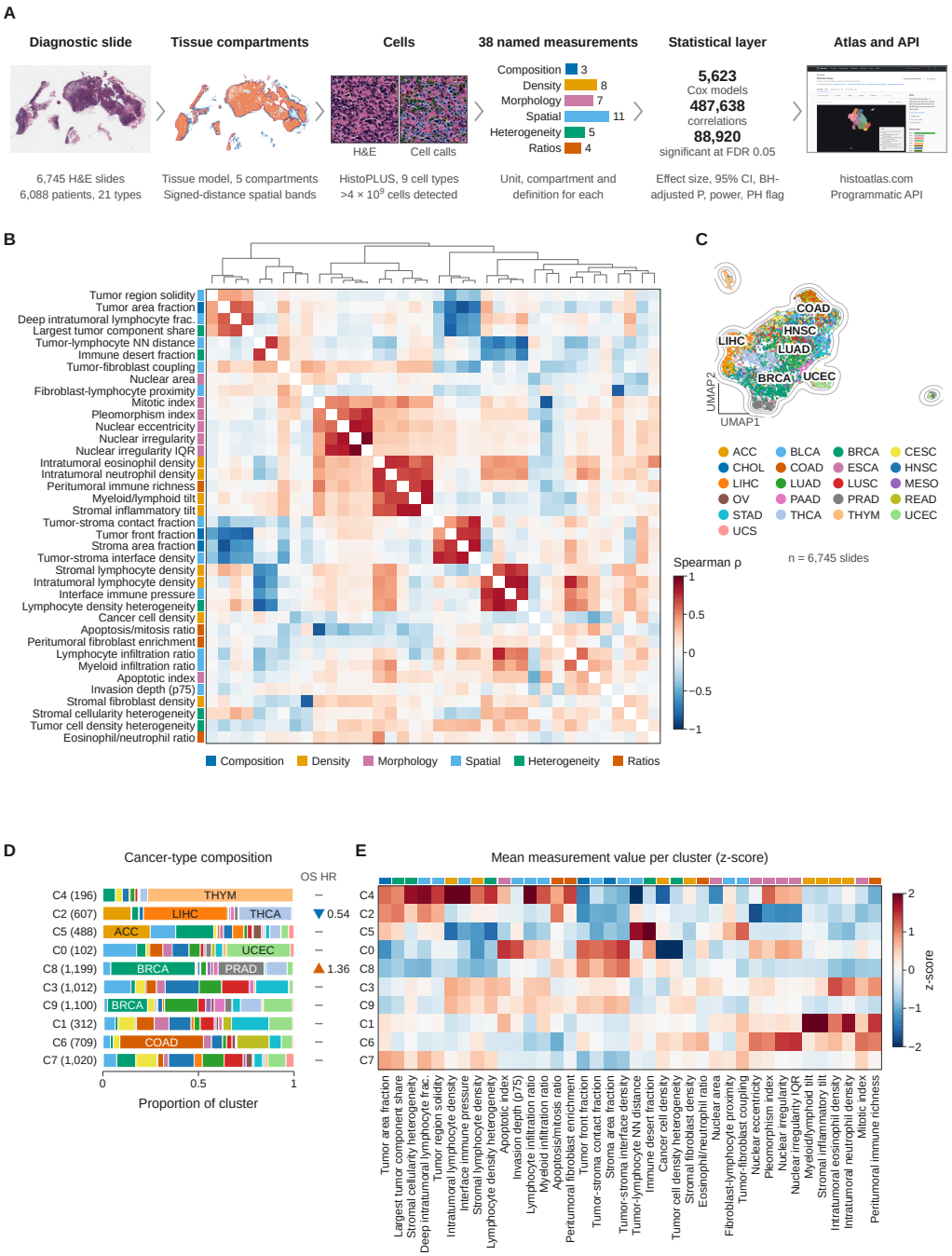

A
Diagnostic slide
Tissue compartments
Cells
38 named measurements
Statistical layer
Atlas and API
H&E
Cell calls
Composition 3
Density 8
Morphology 7
Spatial 11
Heterogeneity 5
Ratios 4
5,623
Cox models
487,638
correlations
88,920
significant at FDR 0.05
6,745 H&E slides
6,088 patients, 21 types
Tissue model, 5 compartments
Signed-distance spatial bands
HistoPLUS, 9 cell types
>4 × 10^9 cells detected
Unit, compartment and definition for each
Effect size, 95% CI, BH-adjusted P, power, PH flag
histoatlas.com
Programmatic API
B
Tumor region solidity
Tumor area fraction
Deep intratumoral lymphocyte frac.
Largest tumor component share
Tumor-lymphocyte NN distance
Immune desert fraction
Tumor-fibroblast coupling
Nuclear area
Fibroblast-lymphocyte proximity
Mitotic index
Pleomorphism index
Nuclear eccentricity
Nuclear irregularity
Nuclear irregularity IQR
Intratumoral eosinophil density
Intratumoral neutrophil density
Peritumoral immune richness
Myeloid/lymphoid tilt
Stromal inflammatory tilt
Tumor-stroma contact fraction
Tumor front fraction
Stroma area fraction
Tumor-stroma interface density
Stromal lymphocyte density
Intratumoral lymphocyte density
Interface immune pressure
Lymphocyte density heterogeneity
Cancer cell density
Apoptosis/mitosis ratio
Peritumoral fibroblast enrichment
Lymphocyte infiltration ratio
Myeloid infiltration ratio
Apoptotic index
Invasion depth (p75)
Stromal fibroblast density
Stromal cellularity heterogeneity
Tumor cell density heterogeneity
Eosinophil/neutrophil ratio
Composition
Density
Morphology
Spatial
Heterogeneity
Ratios
C
COAD
HNSC
LIHC
LUAD
BRCA
UCEC
UMAP2
UMAP1
ACC
BLCA
BRCA
CESC
CHOL
COAD
ESCA
HNSC
LIHC
LUAD
LUSC
MESO
OV
PAAD
PRAD
READ
STAD
THCA
THYM
UCEC
UCS
n = 6,745 slides
Spearman ρ
1
0.5
0
−0.5
−1
D
Cancer-type composition
OS HR
C4 (196) THYM
C2 (607) LIHC THCA
C5 (488) ACC
C0 (102) UCEC
C8 (1,199) BRCA PRAD
C3 (1,012)
C9 (1,100) BRCA
C1 (312)
C6 (709) COAD
C7 (1,020)
0.54
1.36
0
0.5
1
Proportion of cluster
E
Mean measurement value per cluster (z-score)
C4
C2
C5
C0
C8
C3
C9
C1
C6
C7
z-score
2
1
0
−1
−2
Tumor area fraction
Largest tumor component share
Stromal cellularity heterogeneity
Deep intratumoral lymphocyte frac.
Tumor region solidity
Intratumoral lymphocyte density
Interface immune pressure
Stromal lymphocyte density
Lymphocyte density heterogeneity
Apoptotic index
Invasion depth (p75)
Lymphocyte infiltration ratio
Myeloid infiltration ratio
Apoptosis/mitosis ratio
Peritumoral fibroblast enrichment
Tumor front fraction
Tumor-stroma contact fraction
Stroma area fraction
Tumor-stroma interface density
Tumor-lymphocyte NN distance
Immune desert fraction
Cancer cell density
Tumor cell density heterogeneity
Stromal fibroblast density
Eosinophil/neutrophil ratio
Nuclear area
Fibroblast-lymphocyte proximity
Tumor-fibroblast coupling
Nuclear eccentricity
Pleomorphism index
Nuclear irregularity
Nuclear irregularity IQR
Myeloid/lymphoid tilt
Stromal inflammatory tilt
Intratumoral eosinophil density
Intratumoral neutrophil density
Mitotic index
Peritumoral immune richness

Figure 1

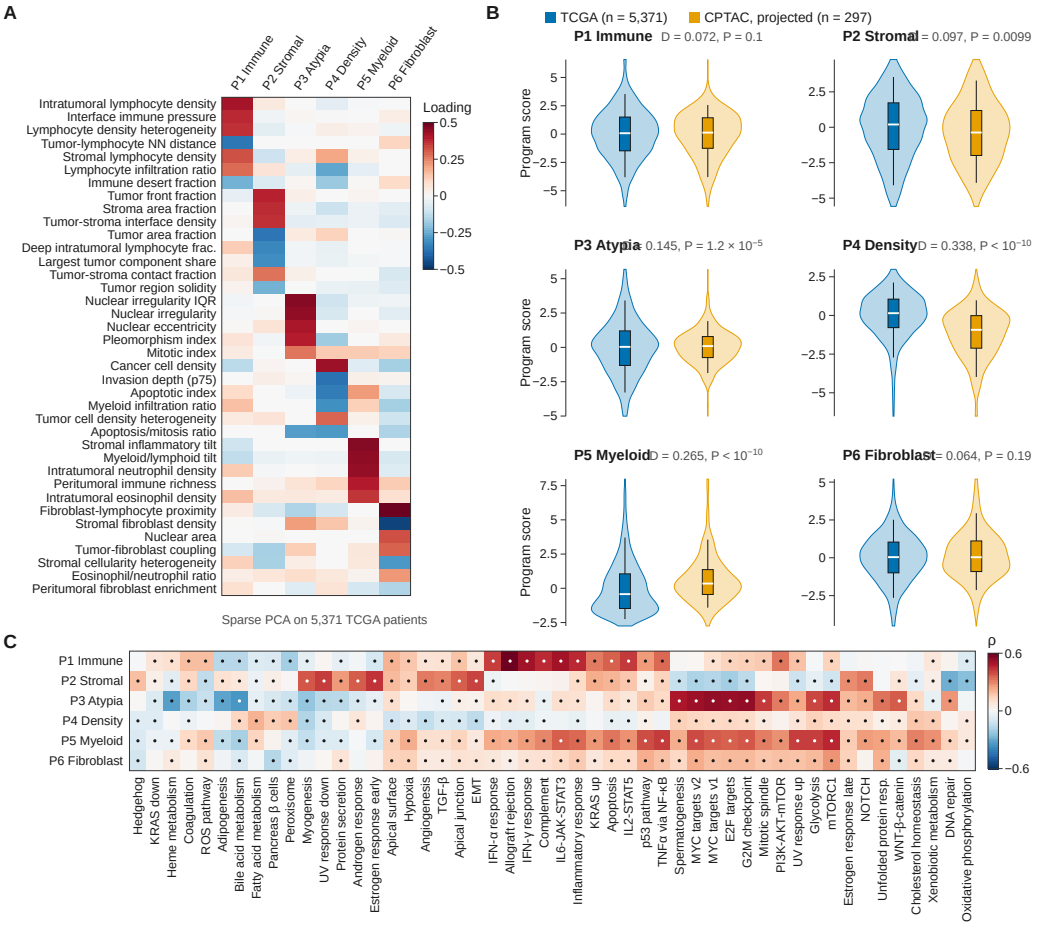

A
P1 Immune
P2 Stromal
P3 Atypia
P4 Density
P5 Myeloid
P6 Fibroblast
Intratumoral lymphocyte density
Interface immune pressure
Lymphocyte density heterogeneity
Tumor-lymphocyte NN distance
Stromal lymphocyte density
Lymphocyte infiltration ratio
Immune desert fraction
Tumor front fraction
Stroma area fraction
Tumor-stroma interface density
Tumor area fraction
Deep intratumoral lymphocyte frac.
Largest tumor component share
Tumor-stroma contact fraction
Tumor region solidity
Nuclear irregularity IQR
Nuclear irregularity
Nuclear eccentricity
Pleomorphism index
Mitotic index
Cancer cell density
Invasion depth (p75)
Apoptotic index
Myeloid infiltration ratio
Tumor cell density heterogeneity
Apoptosis/mitosis ratio
Stromal inflammatory tilt
Myeloid/lymphoid tilt
Intratumoral neutrophil density
Peritumoral immune richness
Intratumoral eosinophil density
Fibroblast-lymphocyte proximity
Stromal fibroblast density
Nuclear area
Tumor-fibroblast coupling
Stromal cellularity heterogeneity
Eosinophil/neutrophil ratio
Peritumoral fibroblast enrichment
Loading
0.5
0.25
0
−0.25
−0.5
Sparse PCA on 5,371 TCGA patients
B
TCGA (n = 5,371)
CPTAC, projected (n = 297)
P1 Immune D = 0.072, P = 0.1
P2 Stromal D = 0.097, P = 0.0099
P3 Atypia D = 0.145, P = 1.2 × 10−5
P4 Density D = 0.338, P < 10−10
P5 Myeloid D = 0.265, P < 10−10
P6 Fibroblast D = 0.064, P = 0.19
Program score
C
P1 Immune
P2 Stromal
P3 Atypia
P4 Density
P5 Myeloid
P6 Fibroblast
Hedgehog
KRAS down
Heme metabolism
Coagulation
ROS pathway
Adipogenesis
Bile acid metabolism
Fatty acid metabolism
Pancreas β cells
Peroxisome
Myogenesis
UV response down
Protein secretion
Androgen response
Estrogen response early
Apical surface
Hypoxia
Angiogenesis
TGF-β
Apical junction
EMT
IFN-α response
Allograft rejection
IFN-γ response
Complement
IL6-JAK-STAT3
Inflammatory response
KRAS up
Apoptosis
IL2-STAT5
p53 pathway
TNFα via NF-κB
Spermatogenesis
MYC targets v2
MYC targets v1
E2F targets
G2M checkpoint
Mitotic spindle
PI3K-AKT-mTOR
UV response up
Glycolysis
mTORC1
Estrogen response late
NOTCH
Unfolded protein resp.
WNT-β-catenin
Cholesterol homeostasis
Xenobiotic metabolism
DNA repair
Oxidative phosphorylation
ρ
0.6
0
−0.6

Figure 2

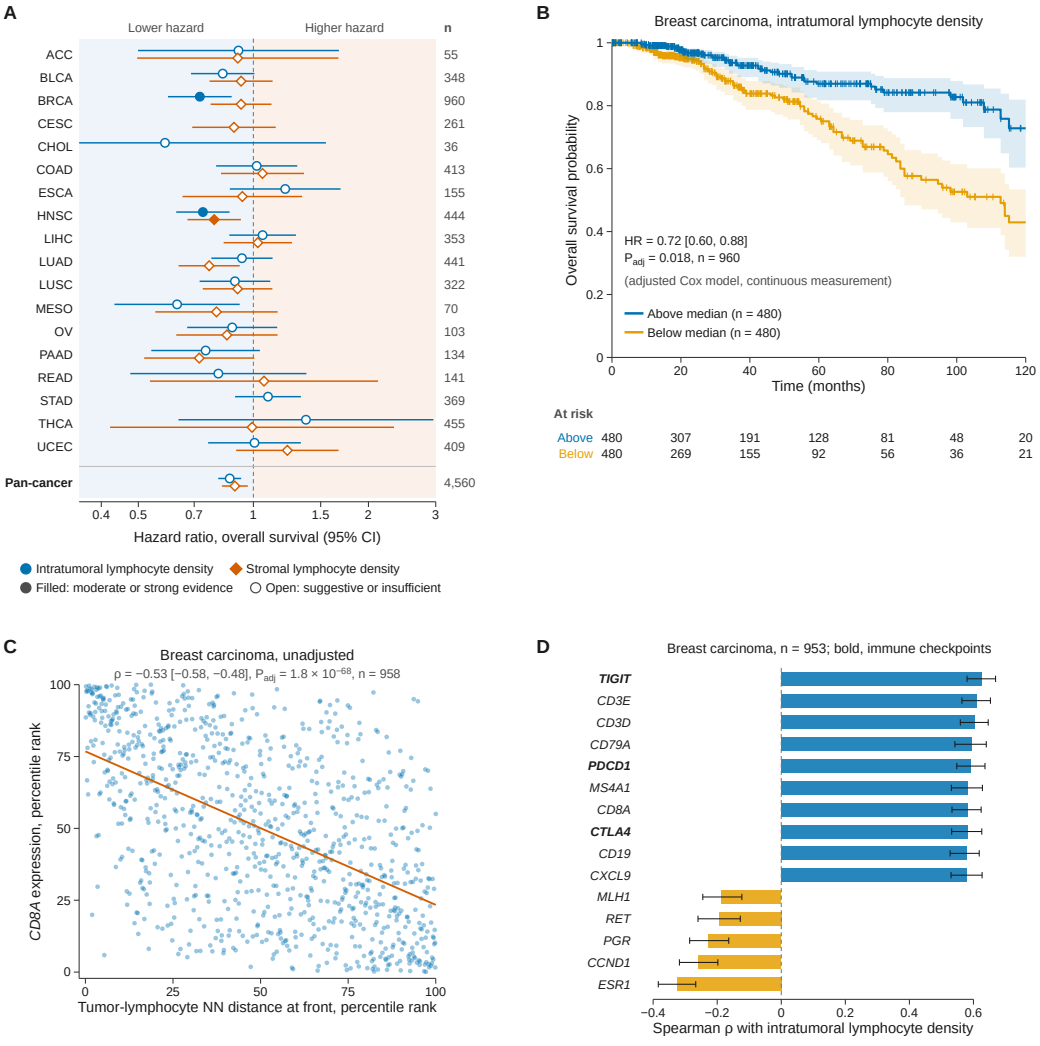

A
Lower hazard
Higher hazard
n
ACC 55
BLCA 348
BRCA 960
CESC 261
CHOL 36
COAD 413
ESCA 155
HNSC 444
LIHC 353
LUAD 441
LUSC 322
MESO 70
OV 103
PAAD 134
READ 141
STAD 369
THCA 455
UCEC 409
Pan-cancer 4,560
0.4 0.5 0.7 1 1.5 2 3
Hazard ratio, overall survival (95% CI)
Intratumoral lymphocyte density
Stromal lymphocyte density
Filled: moderate or strong evidence
Open: suggestive or insufficient
B
Breast carcinoma, intratumoral lymphocyte density
Overall survival probability
HR = 0.72 [0.60, 0.88]
Padj = 0.018, n = 960
(adjusted Cox model, continuous measurement)
Above median (n = 480)
Below median (n = 480)
Time (months)
At risk
Above 480 307 191 128 81 48 20
Below 480 269 155 92 56 36 21
C
Breast carcinoma, unadjusted
ρ = −0.53 [−0.58, −0.48], Padj = 1.8 × 10−68, n = 958
CD8A expression, percentile rank
Tumor-lymphocyte NN distance at front, percentile rank
D
Breast carcinoma, n = 953; bold, immune checkpoints
TIGIT
CD3E
CD3D
CD79A
PDCD1
MS4A1
CD8A
CTLA4
CD19
CXCL9
MLH1
RET
PGR
CCND1
ESR1
Spearman ρ with intratumoral lymphocyte density

Figure 3

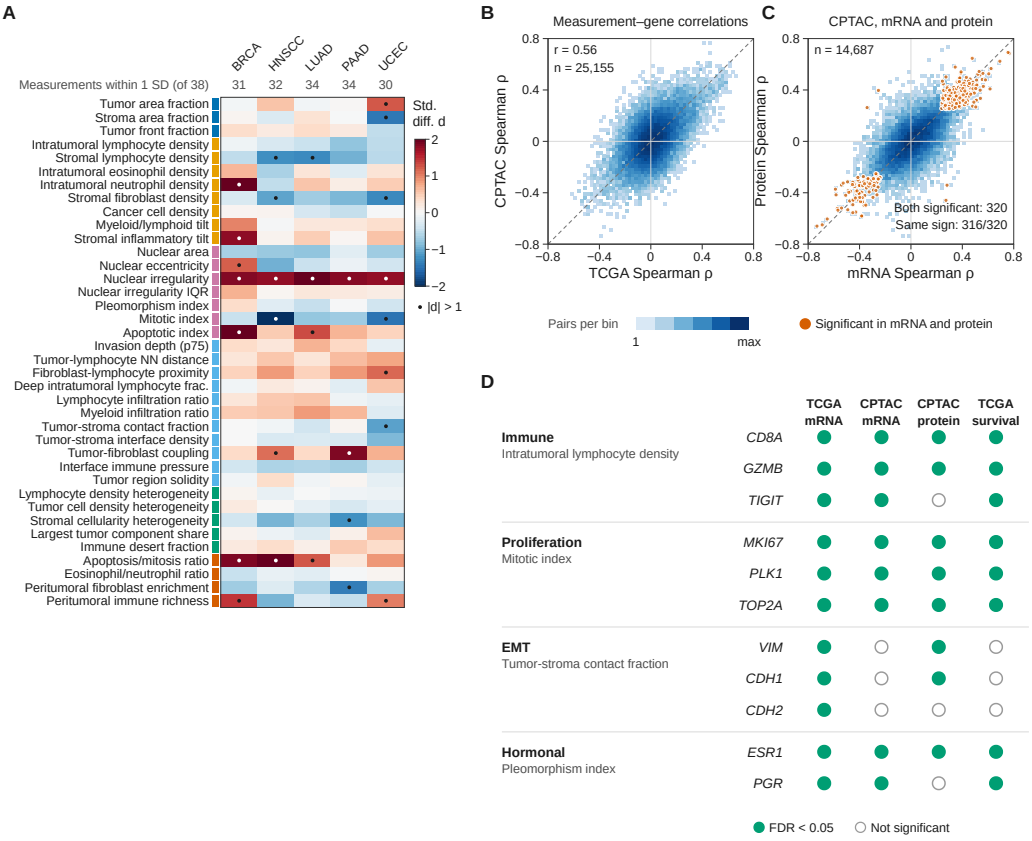

A
BRCA
HNSCC
LUAD
PAAD
UCEC
Measurements within 1 SD (of 38)
31
32
34
34
30
Tumor area fraction
Stroma area fraction
Tumor front fraction
Intratumoral lymphocyte density
Stromal lymphocyte density
Intratumoral eosinophil density
Intratumoral neutrophil density
Stromal fibroblast density
Cancer cell density
Myeloid/lymphoid tilt
Stromal inflammatory tilt
Nuclear area
Nuclear eccentricity
Nuclear irregularity
Nuclear irregularity IQR
Pleomorphism index
Mitotic index
Apoptotic index
Invasion depth (p75)
Tumor-lymphocyte NN distance
Fibroblast-lymphocyte proximity
Deep intratumoral lymphocyte frac.
Lymphocyte infiltration ratio
Myeloid infiltration ratio
Tumor-stroma contact fraction
Tumor-stroma interface density
Tumor-fibroblast coupling
Interface immune pressure
Tumor region solidity
Lymphocyte density heterogeneity
Tumor cell density heterogeneity
Stromal cellularity heterogeneity
Largest tumor component share
Immune desert fraction
Apoptosis/mitosis ratio
Eosinophil/neutrophil ratio
Peritumoral fibroblast enrichment
Peritumoral immune richness
Std. diff. d
2
1
0
−1
−2
• |d| > 1
B
Measurement–gene correlations
r = 0.56
n = 25,155
CPTAC Spearman ρ
TCGA Spearman ρ
0.8
0.4
0
−0.4
−0.8
Pairs per bin
1
max
C
CPTAC, mRNA and protein
n = 14,687
Protein Spearman ρ
mRNA Spearman ρ
Both significant: 320
Same sign: 316/320
Significant in mRNA and protein
D
TCGA mRNA
CPTAC mRNA
CPTAC protein
TCGA survival
Immune
Intratumoral lymphocyte density
CD8A
GZMB
TIGIT
Proliferation
Mitotic index
MKI67
PLK1
TOP2A
EMT
Tumor-stroma contact fraction
VIM
CDH1
CDH2
Hormonal
Pleomorphism index
ESR1
PGR
FDR < 0.05
Not significant

Figure 4

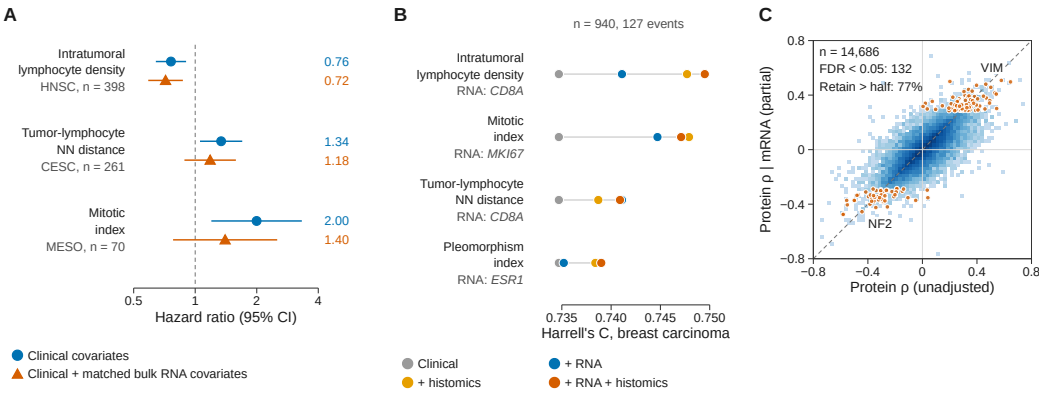

A
Intratumoral
lymphocyte density
HNSC, n = 398
0.76
0.72
Tumor-lymphocyte
NN distance
CESC, n = 261
1.34
1.18
Mitotic
index
MESO, n = 70
2.00
1.40
0.5
1
2
4
Hazard ratio (95% CI)
Clinical covariates
Clinical + matched bulk RNA covariates
B
n = 940, 127 events
Intratumoral
lymphocyte density
RNA: CD8A
Mitotic
index
RNA: MKI67
Tumor-lymphocyte
NN distance
RNA: CD8A
Pleomorphism
index
RNA: ESR1
0.735
0.740
0.745
0.750
Harrell's C, breast carcinoma
Clinical
+ histomics
+ RNA
+ RNA + histomics
C
n = 14,686
FDR < 0.05: 132
Retain > half: 77%
VIM
NF2
0.8
0.4
0
−0.4
−0.8
Protein ρ | mRNA (partial)
Protein ρ (unadjusted)

Figure 5

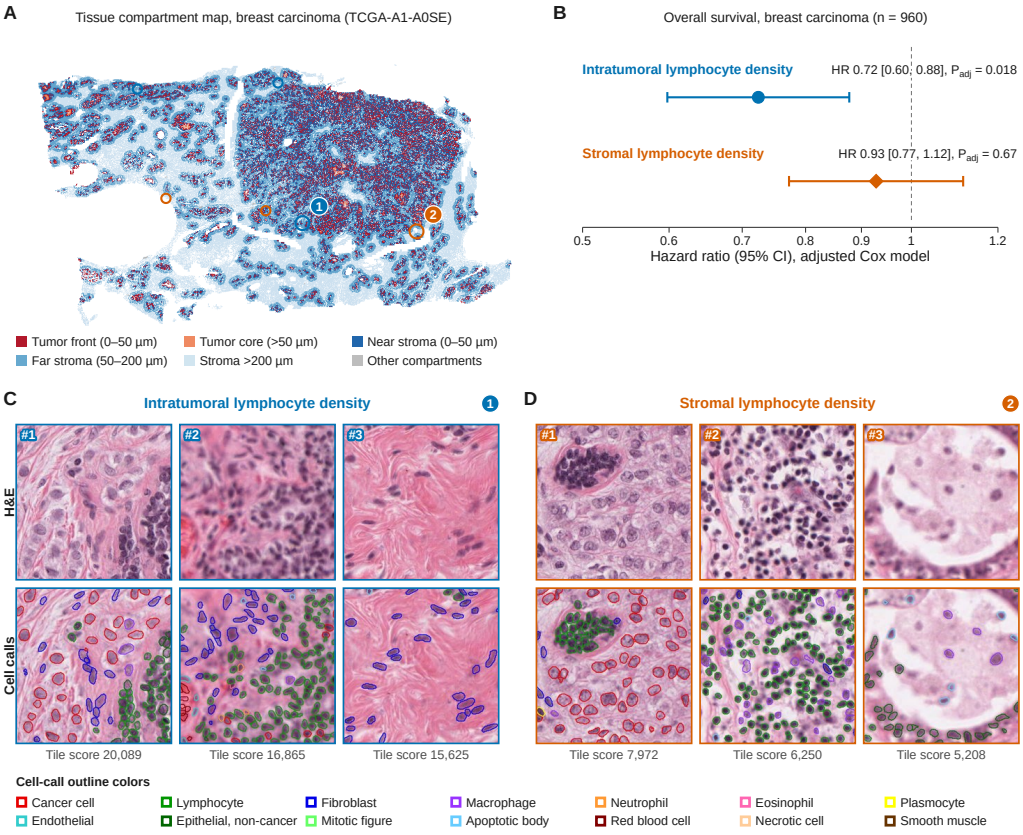

A
Tissue compartment map, breast carcinoma (TCGA-A1-A0SE)
Tumor front (0–50 µm)
Tumor core (>50 µm)
Near stroma (0–50 µm)
Far stroma (50–200 µm)
Stroma >200 µm
Other compartments
B
Overall survival, breast carcinoma (n = 960)
Intratumoral lymphocyte density
HR 0.72 [0.60, 0.88], P adj = 0.018
Stromal lymphocyte density
HR 0.93 [0.77, 1.12], P adj = 0.67
0.5
0.6
0.7
0.8
0.9
1
1.2
Hazard ratio (95% CI), adjusted Cox model
C
Intratumoral lymphocyte density
D
Stromal lymphocyte density
H&E
Cell calls
#1
#2
#3
Tile score 20,089
Tile score 16,865
Tile score 15,625
Tile score 7,972
Tile score 6,250
Tile score 5,208
Cell-call outline colors
Cancer cell
Lymphocyte
Fibroblast
Macrophage
Neutrophil
Eosinophil
Plasmocyte
Endothelial
Epithelial, non-cancer
Mitotic figure
Apoptotic body
Red blood cell
Necrotic cell
Smooth muscle

Figure 6